

Literature resolution: self-adjoint finite-dimensional Lindbladians have positive complete log-Sobolev constant

Automated Math Discovery Run `fa_banach_001`; Model: GPT5.6
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Status: literature already answered (Conjecture 1.4 only). The conjecture in arXiv:2006.14578 that every self-adjoint Lindbladian L satisfies $\text{CLSI}(L) > 0$ follows from Theorem 3.3 of Gao–Rouzé, arXiv:2102.04146. The published source paper explicitly records this later resolution.

Source conjecture

Junge, Li, and LaRacuenta formulate the following as Conjecture 1.4:

$$\text{for every self-adjoint Lindbladian } L, \quad \text{CLSI}(L) > 0.$$

It appears on printed p. 4 of the arXiv source paper [1, p. 4, Conjecture 1.4]. Here “self-adjoint” is with respect to the tracial/GNS inner product, and the ambient algebra in this Lindbladian setting is a finite-dimensional matrix algebra.

Exact later theorem

Answering theorem (Gao–Rouzé, Theorem 3.3). *Let $(P_t = e^{tL})_{t \geq 0}$ be a GNS-symmetric quantum Markov semigroup on a finite-dimensional von Neumann algebra $\mathcal{M} \subseteq B(H)$, and suppose it extends to a semigroup on $B(H)$. If $\mathcal{F}$ is its fixed-point algebra, then it satisfies the complete modified logarithmic Sobolev inequality and*

$$\frac{\lambda(L)}{C_{\tau, \text{cb}}(\mathcal{M} : \mathcal{F})} \leq \alpha_{\text{CMLSI}}(L) \leq 2\lambda(L).$$

This is Theorem 3.3 on printed p. 20 of arXiv:2102.04146 [2, p. 20, Theorem 3.3]. In the source setting $\mathcal{M} = B(H) = M_n$, so the extension hypothesis is automatic. Self-adjointness for the trace is precisely GNS symmetry for the maximally mixed invariant state. Thus the theorem applies to every source Lindbladian and yields the claimed complete inequality; in the usual nontrivial quotient by the fixed algebra the spectral gap is positive by finite-dimensionality. Degenerate fixed directions are already removed by the conditional expectation onto $\mathcal{F}$.

The identification is not merely terminological. The 2025 published version of *Graph Hörmander Systems* explicitly states that Gao and Rouzé subsequently proved the conjectured positivity.

Scope boundary: a different conjecture remains open

The same source also asks for an anti-transference identity recovering $\text{CLSI}(\Delta_X)$ as an infimum of constants of finite-dimensional transferred generators. The published paper proves the corresponding identity for the strengthened CLSI^+ constant but continues to label the plain-CLSI identity a conjecture. Gao–Rouzé’s finite-dimensional positivity theorem does not establish that equality. Accordingly, this packet closes only Conjecture 1.4 and makes no claim that every open problem in arXiv:2006.14578 has been solved.

References

- [1] M. Junge, H. Li, and N. LaRacuenta, *Graph Hörmander Systems*, Ann. Henri Poincaré 26 (2025), 2683–2736; arXiv:2006.14578.
- [2] L. Gao and C. Rouzé, *Complete entropic inequalities for quantum Markov chains*, Arch. Ration. Mech. Anal. 245 (2022), 183–238; arXiv:2102.04146.